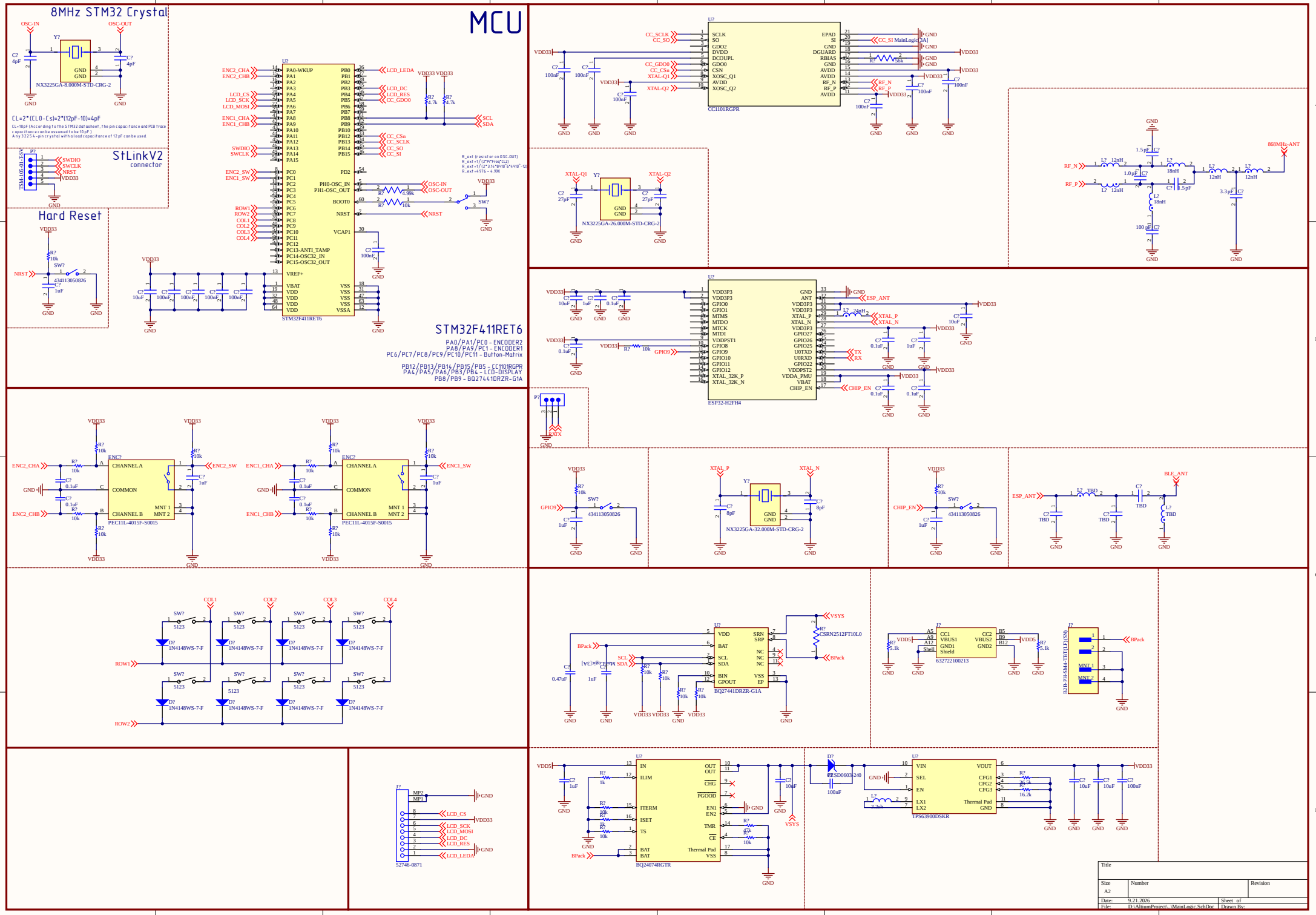

MCU
8MHz STM32 Crystal
OSC-IN
OSC-OUT
NX3225GA-8.000M-STD-CRG-2
CL=2*(CL0-Cs)=2*(12pF-10)=4pF
CL=10pF (According to the STM32 datasheet, the pin capacitance and PCB trace capacitance can be assumed to be 10 pF.)
Any 3225 4-pin crystal with a load capacitance of 12 pF can be used.
StLinkV2
connector
Hard Reset
STM32F411RET6
PA0/PA1/PC0 - ENCODER2
PA6/PA9/PC1 - ENCODER1
PC6/PC7/PC8/PC9/PC10/PC11 - Button-Matrix
PB12/PB13/PB14/PB15/PB5 - CC1101RGPR
PA4/PA5/PA6/PB3/PB4 - LCD-DISPLAY
PB8/PB9 - BQ27441DRZR-G1A
CC1101RGPR
NX3225GA-26.000M-STD-CRG-2
868MHz-ANT
ESP32-H2FH4
ESP_ANT
BLE_ANT
PEC11L-4015F-S0015
1N4148WS-7-F
BQ27441DRZR-G1A
CSRN2512FT10L0
BQ24074RGTR
TPS63900DSKR
632722100213
Title
Size
A2
Number
Revision
Date:
9.21.2026
Sheet of
File:
D:\AltiumProject\...\MainLogic.SchDoc
Drawn By: